Keith's Mock Exam for MA1200 Set 1 Solutions

This document is primarily intended for the uses of students registered for PALSI at CityUHK, and participated in Keith TSE's section.

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Question 1

(a) Let $f(x) = \frac{(3x+2)^5}{x^4}$.
Using the quotient rule, where $u = (3x+2)^5$, $v = x^4$, $u' = 15(3x+2)^4$, $v' = 4x^3$,
$f'(x) = \frac{15(3x+2)^4 \cdot x^4 - (3x+2)^5 \cdot 4x^3}{x^8} = \frac{(3x+2)^4 x^3 [15x - 4(3x+2)]}{x^8} = \frac{(3x+2)^4(3x-8)}{x^5}$.

(b) Let $f(x) = e^{\sin 4x}$.
$f'(x) = e^{\sin 4x} \cdot 4\cos 4x$.

(c) Let $f(x) = \sinh x^4$.
$f'(x) = \cosh x^4 \cdot 4x^3$.

(d) Let $f(x) = \tan^{-1}(4x^2 - x)$.
$f'(x) = \frac{8x-1}{1+(4x^2-x)^2}$.

(e) Let $f(x) = \log_e(\log_e \sin x)$.
The chain: outer ln, inner ln sin x, derivative 1 / (ln sin x) * 1 / sin x * cos x = 1 / (ln sin x) * cot x.
$f'(x) = \frac{\cot x}{\log_e \sin x}$.

Question 2

(a) The function $f(x) = |\cot x|$ for $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ is not defined at $x = 0$ because cot 0 is undefined (sin 0 = 0).
Therefore, f(x) is not differentiable at x = 0, as differentiability requires the function to be defined at the point.

(b) Given $x = \cos t + \log_e \tan\left(\frac{t}{2}\right)$, $y = \sin t$, $0 < t < \pi$.
First, $dx/dt = -\sin t + \csc t = \frac{\cos^2 t}{\sin t}$. dy/dt = cos t.
Thus, $\frac{dy}{dx} = \frac{\cos t}{\frac{\cos^2 t}{\sin t}} = \tan t$.
Now, $\frac{d}{dt}\left(\frac{dy}{dx}\right) = \sec^2 t$.
So $\frac{d^2y}{dx^2} = \frac{\sec^2 t}{\frac{\cos^2 t}{\sin t}} = \frac{\sin t}{\cos^4 t}$.

Question 3

(a) (i)The Taylor series expansion of $\cos x$ around $x = 0$ is $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + O(x^8)$.
The numerator is $2 - 2x^3 - 2\cos x = 2 - 2x^3 - 2\left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + O(x^8)\right) = 2 - 2x^3 - 2 + x^2 - \frac{x^4}{12} + \frac{x^6}{360} + O(x^8) = x^2 - 2x^3 - \frac{x^4}{12} + \frac{x^6}{360} + O(x^8)$.
Dividing by x^5: $\frac{x^2 - 2x^3 - \frac{x^4}{12} + \frac{x^6}{360} + O(x^8)}{x^5} = \frac{1}{x^3} - \frac{2}{x^2} - \frac{1}{12x} + \frac{x}{360} + O(x^3)$.
As $x \to 0^+$, this approaches $+\infty$; as $x \to 0^-$, this approaches $-\infty$.
Thus, the limit does not exist.

(ii) Let $h = x - 1$, so as $x \to 1$, $h \to 0$.
The limit becomes $\lim_{h\to 0} h \cot\left(\frac{\pi(1+h)}{2}\right) = \lim_{h\to 0} h \cot\left(\frac{\pi}{2} + \frac{\pi h}{2}\right) = \lim_{h\to 0} h\left(-\tan\frac{\pi h}{2}\right) = -\lim_{h\to 0} h \tan\frac{\pi h}{2}$.
Since $\tan\theta \approx \theta$ as $\theta \to 0$, $\tan\frac{\pi h}{2} \approx \frac{\pi h}{2}$, so $-h \cdot \frac{\pi h}{2} = -\frac{\pi}{2}h^2 \to 0$.
Thus, the limit is 0.

(b) By first principles, $\frac{d}{dx}(\cos x) = \lim_{h\to 0} \frac{\cos(x+h) - \cos x}{h}$.
Using the hint: $\cos(x+h) - \cos x = -2\sin\left(\frac{(x+h)+x}{2}\right)\sin\left(\frac{(x+h)-x}{2}\right) = -2\sin\left(x + \frac{h}{2}\right)\sin\left(\frac{h}{2}\right)$.
So, $\frac{\cos(x+h)-\cos x}{h} = \frac{-2\sin\left(x+\frac{h}{2}\right)\sin\left(\frac{h}{2}\right)}{h} = -\sin\left(x + \frac{h}{2}\right) \cdot \frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}}$.
As $h \to 0$, this approaches $-\sin x \cdot 1 = -\sin x$.

Question 4

(a) The perimeter P_n of a regular n-sided polygon inscribed in a unit circle (radius 1) is given by the formula $P_n = 2n\sin\left(\frac{\pi}{n}\right)$.
For n = 5:
$P_5 = 2 \times 5 \times \sin\left(\frac{\pi}{5}\right) = 10\sin\left(\frac{\pi}{5}\right) \approx 5.878$
For n = 10:
$P_{10} = 2 \times 10 \times \sin\left(\frac{\pi}{10}\right) = 20\sin\left(\frac{\pi}{10}\right) \approx 6.180$
For n = 20:
$P_{20} = 2 \times 20 \times \sin\left(\frac{\pi}{20}\right) = 40\sin\left(\frac{\pi}{20}\right) \approx 6.257$
For n = 40:
$P_{40} = 2 \times 40 \times \sin\left(\frac{\pi}{40}\right) = 80\sin\left(\frac{\pi}{40}\right) \approx 6.277$

(b) $\frac{64P_{40} - 32P_{20} + 4P_{10} - P_5}{42} \approx 5.25$

Question 5

(a) $y' = qb(bx+c)^{q-1}$.
$y'' = q(q-1)b^2(bx+c)^{q-2}$.
$y''' = q(q-1)(q-2)b^3(bx+c)^{q-3}$.
By pattern, the nth derivative is $\frac{d^n y}{dx^n} = q(q-1)\cdots(q-n+1)b^n(bx+c)^{q-n}$.

(b) Assume $\frac{5x^2}{(2x-1)^3} = \frac{A}{2x-1} + \frac{B}{(2x-1)^2} + \frac{C}{(2x-1)^3}$.
Multiply both sides by $(2x-1)^3$: $5x^2 = A(2x-1)^2 + B(2x-1) + C$.
Expand: $A(4x^2 - 4x + 1) + B(2x-1) + C = 4Ax^2 + (-4A+2B)x + (A - B + C)$.
Equate coefficients:
For x^2: $4A = 5 \Rightarrow A = 5/4$. For x: $-4A + 2B = 0 \Rightarrow -5 + 2B = 0 \Rightarrow B = 5/2$.
For constant: $A - B + C = 0 \Rightarrow 5/4 - 5/2 + C = 0 \Rightarrow 5/4 - 10/4 + C = 0 \Rightarrow C = 5/4$.
Thus, $\frac{5x^2}{(2x-1)^3} = \frac{\frac{5}{4}}{2x-1} + \frac{\frac{5}{2}}{(2x-1)^2} + \frac{\frac{5}{4}}{(2x-1)^3}$.

(c) The t-th derivative of $\frac{A}{2x-1}$ is $A(-1)^t t! 2^t (2x-1)^{-(t+1)}$.
The t-th derivative of $\frac{B}{(2x-1)^2}$ is $B(-1)^t (t+1)! 2^t (2x-1)^{-(t+2)}$.
The t-th derivative of $\frac{C}{(2x-1)^3}$ is $C(-1)^t \frac{(t+2)!}{2!} 2^t (2x-1)^{-(t+3)}$.
Thus, $\frac{d^t}{dx^t}\left(\frac{5x^2}{(2x-1)^3}\right) = \frac{5}{4}(-1)^t 2^t t!(2x-1)^{-(t+1)} + \frac{5}{2}(-1)^t 2^t (t+1)!(2x-1)^{-(t+2)} + \frac{5}{4}(-1)^t 2^t \frac{(t+2)!}{2}(2x-1)^{-(t+3)}$.

Question 6

(a) $f(-x) = [(-x)^2]^{1/3} - [(-x)^2 - 9]^{1/3} = (x^2)^{1/3} - (x^2-9)^{1/3} = f(x)$
Therefore, it is even.
For $x > 0$:
If $x > 3$, $x^2 - 9 > 0$, and $x^2 > x^2 - 9$, so $(x^2)^{1/3} > (x^2-9)^{1/3}$, hence $f(x) > 0$.
If $0 < x < 3$, $x^2 - 9 < 0$, so $(x^2-9)^{1/3} < 0$, hence $f(x) = (x^2)^{1/3} - (negative) > 0$.
If $x = 3$, $f(3) = (9)^{1/3} - 0 > 0$.

(b) Let $a = x^3$, $b = (x^2-3)^{1/3}$.
Then $(a-b)(a+b) = a^2 - b^2 = x^6 - [(x^2-3)^{1/3}]^2 = x^6 - (x^2-3)^{2/3}$.

(c) $f(x) = (x^2)^{1/3}(1 - (1 - 9/x^2)^{1/3})$.
Using binomial expansion, $(1-9/x^2)^{1/3} = 1 - 3/x^2 + O(1/x^4)$. So $1 - (1 - 3/x^2 + O(1/x^4)) = 3/x^2 - O(1/x^4)$.
Then $f(x) = x^{2/3} \cdot 3/x^2 + O(x^{2/3-4}) = 3x^{-4/3} + O(x^{-10/3}) \to 0$ as $x \to +\infty$.

(d) $f'(x) = \frac{2x}{3}\left[(x^2)^{-2/3} - (x^2-9)^{-2/3}\right]$.
Set $f'(x) = 0$.
Either $x = 0$, or $(x^2)^{-2/3} = (x^2-9)^{-2/3}$.
For the second, $x^{4/3} = (x^2-9)^{2/3}$.
Raise to 3/2: $x^2 = x^2 - 9$, 0 = -9, impossible.
At x = 0, the derivative is not defined or not zero (limits to $\pm\infty$ from sides). Thus, no real solutions.

Question 7

(a) The hyperbola is $\frac{x^2}{4} - y^2 = 1$, so $a^2 = 4$, $a = 2$, $b^2 = 1$, $b = 1$.
Vertices: $(\pm a, 0) = (\pm 2, 0)$.
Foci: $(\pm c, 0)$, where $c = \sqrt{a^2 + b^2} = \sqrt{4 + 1} = \sqrt{5}$.

(b) Let $x = t + t^{-1}$, $y = \frac{t - t^{-1}}{2}$.
Then $x^2 = (t + t^{-1})^2 = t^2 + 2 + t^{-2}$.
$\frac{x^2}{4} = \frac{t^2+2+t^{-2}}{4}$.
$y^2 = \left(\frac{t-t^{-1}}{2}\right)^2 = \frac{(t-t^{-1})^2}{4} = \frac{t^2-2+t^{-2}}{4}$.
$\frac{x^2}{4} - y^2 = \frac{t^2+2+t^{-2}}{4} - \frac{t^2-2+t^{-2}}{4} = \frac{4}{4} = 1$.
Thus, P lies on H.

(c) The equation of the tangent at (x_0, y_0) is $\frac{xx_0}{4} - yy_0 = 1$.
Substitute $x_0 = t + t^{-1}$, $y_0 = \frac{t-t^{-1}}{2}$:
$\frac{x(t+t^{-1})}{4} - y \cdot \frac{t-t^{-1}}{2} = 1$.

Question 8

(a) Let $u = 3\log_e(1 + x)$, so $y = \cos u$.
Then $\frac{dy}{dx} = -\sin u \cdot \frac{3}{1+x}$.
Then $\frac{d^2y}{dx^2} = -\left[\cos u \cdot \frac{3}{1+x} \cdot \frac{1}{1+x} - \sin u \cdot \frac{1}{(1+x)^2}\right] \cdot 3 = -3\left[\frac{3\cos u}{(1+x)^2} - \frac{\sin u}{(1+x)^2}\right] = \frac{-9\cos u + 3\sin u}{(1+x)^2}$.
Then $(1 + x)^2 \frac{d^2y}{dx^2} = -9\cos u + 3\sin u$.
$(1 + x)\frac{dy}{dx} = (1 + x)\left(-\frac{3\sin u}{1+x}\right) = -3\sin u$.
Sum: $-9\cos u + 3\sin u - 3\sin u = -9\cos u$.
Then $+9y = -9\cos u + 9\cos u = 0$.

(b) The equation for n=0 is $(1 + x)^2 \frac{d^2y}{dx^2} + (1)(1 + x)\frac{dy}{dx} + (0 + 9)(1 + x)^0 y = 0$. Differentiate using Leibniz rule to obtain higher forms.
Differentiate once: $2(1 + x)\frac{d^2y}{dx^2} + (1 + x)^2 \frac{d^3y}{dx^3} + \frac{dy}{dx} + (1 + x)\frac{d^2y}{dx^2} + 9\frac{dy}{dx} = 0$. Simplify: $(1 + x)^2 \frac{d^3y}{dx^3} + 3(1 + x)\frac{d^2y}{dx^2} + 10\frac{dy}{dx} = 0$.
This is the form for n=1 with powers adjusted by multiplying by (1 + x) if needed, but the pattern holds.
By repeated differentiation and applying Leibniz rule to each product term, the general form is deduced for n = 1, 2, ...

(c) At $x = 0$: $y(0) = 1$, $\frac{dy}{dx}(0) = 0$, $\frac{d^2y}{dx^2}(0) = -9$.
From the n=1 equation at x=0: $\frac{d^3y}{dx^3} + 3(-9) + 10(0) = 0$, so $\frac{d^3y}{dx^3} = 27$.
From the n=2 equation at x=0: $\frac{d^4y}{dx^4} + 5(27) + 13(-9) = 0$, so $\frac{d^4y}{dx^4} = -135 + 117 = -18$.
The series is $1 + 0 \cdot x + \frac{-9}{2}x^2 + \frac{27}{6}x^3 + \frac{-18}{24}x^4 = 1 - \frac{9}{2}x^2 + \frac{9}{2}x^3 - \frac{3}{4}x^4$.